

Profibus

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Abstract—PROFIBUS is a widely used industrial fieldbus protocol that enables deterministic communication between controllers, sensors, actuators, and distributed field devices. This paper presents a technical overview of PROFIBUS, including its architecture and main variants. It also discusses key limitations of PROFIBUS in modern industrial automation. Furthermore, PROFIBUS is compared with modern Industrial Ethernet protocols. Therefore, gradual migration through gateways, proxies, and hybrid architectures is identified as the most practical approach for combining existing PROFIBUS infrastructure with modern industrial communication technologies.

Index Terms—PROFIBUS, PROFIBUS-DP, PROFIBUS-PA, fieldbus, Industrial Ethernet, PROFINET, master-slave token passing, deterministic communication, intrinsic safety

I. INTRODUCTION

The evolution of the Distributed Control System (DCS) has shifted industrial automation from centralized control to distributed architectures, driven by the inherent need to reduce costs, improve product quality, and increase system flexibility [1], [2], . Early systems relied on analog point-to-point wiring, such as the 4–20 mA current loop, which required individual physical connections for every sensor and actuator. This approach scales poorly as plant complexity grows. The transition to digital fieldbus communication solved these issues. By allowing multiple devices to share a single communication cable, fieldbus technologies enabled the decentralization of control functions, allowed for the direct integration of intelligent field devices, and reduced wiring complexity across industrial plants [2], [3].

PROFIBUS emerged as a leading standard during this transition to decentralized control. It operates primarily on an RS-485 physical layer and utilizes a deterministic master-slave polling mechanism [4]. In this architecture, active stations (masters) control the bus communication, while passive field devices (slaves) respond to specific data requests. PROFIBUS ensures predictable cycle times by separating time-critical cyclic data exchange from acyclic configuration and diagnostic tasks [2], [4]. Because of this predictable timing and reliable data delivery, it became the standard for time-critical applications ranging from basic process automation to high-speed manufacturing and motion control [5].

The shift toward Industry 4.0 and the deployment of cyber-physical systems however, have introduced new operational demands that exceed the capabilities of traditional serial networks [6]. Modern automation architectures require the convergence of Information Technology (IT) and Operational Technology (OT). Furthermore, the integration of advanced

edge computing hardware, such as deploying specialized accelerators for multi-camera inference and real-time quality control directly on the production line—demands substantial data throughput. These data-intensive applications expose the inherent bandwidth limits of traditional fieldbus technologies like PROFIBUS, which typically maxes out at 12 Mbps [7], [8].

As a result, while PROFIBUS remains heavily used in legacy systems due to its established reliability, the industry is transitioning toward Industrial Ethernet solutions. Technologies such as PROFINET and EtherCAT offer the higher bandwidth, scalable topologies, and seamless IT integration required to support modern, data-driven manufacturing environments, making them more suited for future-proofing [6], [8], [9].

To analyze this technological shift, this paper provides a technical evaluation of PROFIBUS. It examines the protocol’s communication architecture, timing behavior, and real-time performance limitations. Furthermore, this study compares these findings against modern Industrial Ethernet architectures, including PROFINET, to quantify the differences in key performance metrics including latency, determinism, and overall cycle time.

II. LITERATURE REVIEW

Prior to the widespread adoption of digital networks in the 1980s, industrial automation relied heavily on centralized control architectures utilizing analog point-to-point wiring (predominantly 4–20 mA analog signals). This traditional method presented significant challenges as manufacturing facilities grew in scale, because each and every sensor and actuator required a dedicated physical cable routed back to a central controller, plant wiring would become highly complex, expensive to install, and difficult to maintain. Additionally, analog interfaces were limited to unidirectional signal transmission, meaning they could not provide advanced device diagnostics, remote configuration, or fault detection [1], [2].

To overcome these physical and functional limitations, a standardized, digital, and decentralized communication network was required. In 1987, a collective of German companies and research institutes initiated a joint project to develop an open fieldbus standard, which led to the creation of PROFIBUS (Process Field Bus) [10], [11]. By utilizing a shared serial bus topology, PROFIBUS allowed multiple intelligent field devices to communicate over a single network cable. This transition from analog to digital communication

resolved the previous scalability issues by drastically reducing wiring costs, enabling bidirectional data exchange, and allowing for comprehensive system diagnostics [2], [12].

To address the diverse requirements of industrial environments, PROFIBUS evolved through three primary versions. The initial version, PROFIBUS FMS (Fieldbus Message Specification), was released in 1989. FMS was designed for complex, non-deterministic communication between controllers and computers at the higher cell level of automation. However, its heavy protocol overhead made it too slow and overly complex for real-time field device control [13], [14].

To resolve the performance bottlenecks of FMS, PROFIBUS DP (Decentralized Peripherals) was introduced in 1993. DP was highly optimized for fast, deterministic, cyclic data exchange between a master controller and remote I/O slaves. Due to its simplicity, speed, and reliability, DP quickly became the industry standard and the most widely installed version of PROFIBUS [13], [15]. Shortly thereafter, PROFIBUS PA (Process Automation) was developed specifically for hazardous and explosive environments. PA utilizes a specialized physical layer that delivers both power and digital communication over the same two-wire cable while restricting current to ensure intrinsic safety [3], [16].

Today, PROFIBUS maintains a massive global installed base across distinct industrial sectors based on these protocol variants. PROFIBUS DP is utilized extensively in factory automation, discrete manufacturing, and motion control, where high-speed reaction times are essential [12], [15]. In contrast, PROFIBUS PA is heavily deployed in process automation industries, such as chemical manufacturing, oil and gas, and wastewater treatment, where explosive risks are present and the monitoring of continuous variables is required [3], [6].

III. PROFIBUS ARCHITECTURE AND OPERATION

This section analyzes how PROFIBUS operates, following the ISO/OSI reference model and then examining the bus access mechanism, communication services, and the three protocol variants (DP, PA, and FMS) in turn.

A. Architecture and the ISO/OSI Reference Model

PROFIBUS uses a master-slave communication model and supports linear-bus, tree, and star topologies. Interoperability between devices from different manufacturers is achieved through standardized device description files (GSD and EDDL) [6]. PROFIBUS is divided into three variants: PROFIBUS-DP for decentralized control and I/O communication, PROFIBUS-PA for process automation, and PROFIBUS-FMS for peer-to-peer communication, although FMS is now largely obsolete [17].

The PROFIBUS technology modules are designed with reference to the ISO/OSI reference model, which defines all elements, structures, and tasks required for communication across seven layers. PROFIBUS deliberately simplifies this model and implements only layers 1, 2, and 7, adding a user-oriented layer above layer 7 that is not itself part of the OSI model [18]. Layer 1 (the physical layer) defines the

transmission medium and bus interface; PROFIBUS supports copper-wire variants, fibre-optic transmission for interference-prone environments and extended distances, and intrinsically safe transmission. Layer 2 (the data link layer, the Fieldbus Data Link, FDL) defines the bus access method and data security through a combination of the master-slave method and the token method. Layer 7 (the application layer) forms the interface to the application; for PROFIBUS-DP the DP communication protocol resides here, whereas for PROFIBUS-FMS the application layer consists of the Fieldbus Message Specification (FMS), the Lower Layer Interface (LLI), and the Fieldbus Management services for layer 7 (FMA 7) [18], [19].

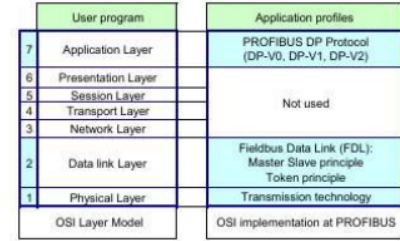


Fig. 1. References between OSI model and PROFIBUS. Reprinted from [18]

B. Bus Access: the Hybrid Master-Slave and Token Mechanism

The PROFIBUS bus access method is a hybrid procedure that combines a token-passing procedure for communication between complex stations (masters) with an underlying master-slave procedure for communication with low-cost peripheral devices (slaves). This hybrid scheme makes PROFIBUS a deterministic network: each station is guaranteed bus access within a fixed time, a property administered by a logical token ring at the data link level. The token passes between the masters, and the station that currently holds the token controls the bus; while holding the token, a master communicates with all of its slaves, either sending data to a slave or requesting data from it, and a slave may transmit only in response to a request from its master. The principle of data exchange is therefore strictly request-response: the master obtains a data telegram from a slave after sending a request telegram, and when the master sends a data telegram to a slave, the slave answers with a status telegram [19].

In multi-master configurations the token forms a logical ring among the master stations: each master is assigned an address, and the next holder of the token is determined by the address values, so that the token performs a polling transfer between the primary stations according to their link tables [19]. Bus timing is governed by the target token rotation time (TTR), actual token rotation time (TRR), and token-holding time (TTH) which is the difference between the former two times. High-priority messages are transmitted first, while low-priority messages are sent only when sufficient time remains within the target rotation period [20].

PROFIBUS defines two classes of master. A class 1 master (DPM1) is a typical central controller, such as a PLC or a DCS

automation station, that exchanges process data cyclically with its assigned slaves; a class 2 master (DPM2) is an engineering or operator station (for example SIMATIC PDM) used for commissioning, configuration, and diagnostics, and need not be permanently connected to the bus. Any device that is neither a class 1 nor a class 2 master is a slave: a passive node, typically a field device such as remote I/O, a drive, a valve, a transducer, or an analyzer, that responds only when addressed. Slave devices are further distinguished as compact devices, with a fixed set of input and output data, or modular devices, which comprise a head station and a number of slots whose modules, frequently virtual modules existing only in the device software, can be combined flexibly to match the required process data [18].

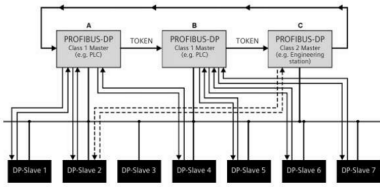


Fig. 2. Master and Slave classes and communication. Reprinted from [18]

C. Communication Services: Cyclic and Acyclic Exchange

PROFIBUS distinguishes cyclic and acyclic communication services. Cyclic services carry time-critical process I/O data and are executed every bus cycle, typically by a class 1 master. Acyclic services are initiated on demand and are used for configuration, parameterization, diagnostics, and alarm handling. Since acyclic communication is not time-critical, completion may require multiple bus cycles [18].

D. PROFIBUS-DP (Decentralized Peripherals)

PROFIBUS-DP is the core variant of PROFIBUS and provides the fundamental functions used by other versions. It is designed for fast, decentralized control of field devices such as PLCs, distributed I/O, and drives [7]. Communication follows a master–slave model where an active master (PLC or controller) cyclically exchanges data with passive devices. The master sends output data (e.g., drive setpoints), and the slave responds with input data (e.g., sensor measurements) [21].

DP typically runs on RS-485 due to its speed and noise immunity, but can also use fibre optics or wireless links [18]. Its support for data rates up to 12 Mbps and payloads up to 244 bytes allows complete I/O updates to be transferred in a single cycle, enabling fast and efficient real-time control in both factory and process automation [18].

E. Functional Levels: DP-V0, DP-V1, and DP-V2

The PROFIBUS-DP protocol is divided into three performance levels [18].

DP-V0 provides the basic communication function, comprising cyclic data exchange between a controller and its slave

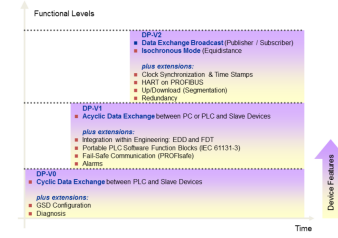


Fig. 3. PROFIBUS DP protocol. Reprinted from [21]

devices together with GSD-based configuration and device-, module-, and channel-specific diagnostics for fast fault localisation, such as “excess temperature” or “short circuit on output” [21]. The format of the exchanged data depends on the device profile.

DP-V1 supplements DP-V0 with functions for acyclical communication, i.e. for functions such as parameterization, operation, monitoring and alarm handling. DP-V1 enables online access to bus nodes via engineering tools for this purpose [21].

Version DP-V2 contains additional functions as extensions of DP-V1, in particular functions which are required for drive control. These include functions for communication between slaves, cycle synchronization and time stamping. Field devices for process automation are typically slaves of performance level DP-V1 and can therefore communicate acyclically to set device parameters [21].

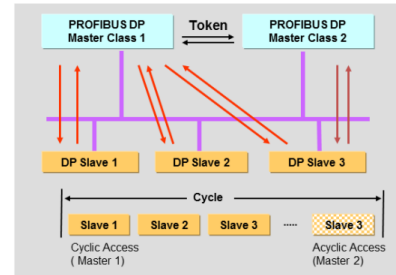


Fig. 4. Cyclical and acyclical communication with DP-V1. Reprinted from [21]

F. PROFIBUS-PA (Process Automation)

PROFIBUS-PA is intended for process automation and provides intrinsically safe operation, long cable lengths, and bus-powered field devices. The PA profile standardizes device functions, parameters, diagnostics, and cyclic process-data formats, improving interoperability and simplifying engineering [19], [21], [22]. Power and communication are carried on the same two-wire cable, reducing installation costs [22].

PROFIBUS-PA uses the same protocol as DP but operates on a different physical layer called MBP (Manchester Coded, Bus Powered). This technology allows both power and data to be transmitted over a single twisted-pair Type A fieldbus cable, reducing wiring complexity and installation cost. It runs at a fixed 31.25 kbps using Manchester II coding with a ± 9

mA signal amplitude, which is sufficient for slow-changing process variables in process automation [17], [23]. A single segment can support up to 32 devices over distances of up to 1,900 m, with field devices requiring at least 9 V DC and 10 mA to operate on the bus [21].

Since PROFIBUS-PA cannot operate independently, it is implemented as a subsystem within a PROFIBUS-DP network, requiring a coupler or link between PA and DP segments. A coupler simply forwards signals and makes PA devices appear as DP stations, but fixes the DP baud rate, whereas a link is an intelligent gateway with its own DP address, allowing flexible DP baud rates and more independent operation [19], [24], [25]. When integrated into higher-level systems such as PROFINET, the controller acts as a gateway connecting PA devices to Ethernet-based networks for centralized control [17].

The PA profile standardizes process-data representation. Analog variables are transmitted as four-byte IEEE floating-point values with an additional status byte, while digital variables use one data byte and one status byte [18].

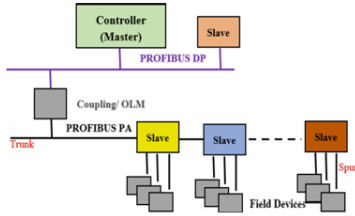


Fig. 5. Industrial network. Reprinted from [23]

Hazardous-area installations are commonly implemented using either the FISCO (Fieldbus Intrinsically Safe Concept) model or the High-Power Trunk concept. FISCO simplifies intrinsic-safety verification by requiring certified components that satisfy predefined electrical limits, while the High-Power Trunk concept separates intrinsically safe spurs from a higher-power trunk cable [26]. Typical PA segments support cable lengths approaching 1900 m and approximately 32 devices per segment [17], [27], [28].

G. PROFIBUS-FMS (Fieldbus Message Specification)

PROFIBUS-FMS is an object-oriented protocol intended for cell-level communication between intelligent controllers, emphasizing functionality and acyclic exchange of complex data rather than fast response times [29], [30]. Its application layer uses the Fieldbus Message Specification (FMS) and Lower Layer Interface (LLI), supporting both dynamically established and predefined communication relationships [6], [29]. Communication relationships are managed through a Communication Relationship List (CRL), while application profiles ensure interoperability between vendors [29]. Due to its high protocol overhead, lower effective speed, and reduced determinism, FMS has largely disappeared from industrial use, leaving DP and PA as the dominant PROFIBUS variants [6].

IV. LIMITATIONS AND IMPROVEMENTS

While PROFIBUS remains a mature and deterministic fieldbus, the demands of Industry 4.0 have exposed a number of structural limitations. This section pairs each limitation with an improvement that has been proposed or demonstrated in the literature, so that the practical headroom of the technology can be assessed.

A. System Integration and Maintenance Complexity

As automation systems expand, integrating and maintaining PROFIBUS systems containing hundreds or thousands of devices becomes more difficult, and equipment ageing raises failure rates over long service lives [6].

Improvement: An improvement combines a PROFIBUS/PROFINET gateway with a unified engineering and configuration environment, IoT-based collection of device data for cloud analysis, an integrated energy-management module for real-time monitoring and automatic parameter adjustment, and a predictive-maintenance strategy that anticipates equipment failure from data analysis so that components can be repaired or replaced in advance, reducing downtime and improving overall efficiency [6].

B. Competition from Emerging Industrial Protocols

Driven by Industry 4.0 and intelligent manufacturing, emerging Ethernet-based protocols such as PROFINET, EtherCAT, and Modbus TCP offer higher data rates, richer topologies, and greater data-processing capability, against which PROFIBUS may not be fast or flexible enough to meet the latest automation needs [6].

Improvement: Ethernet-APL (IEEE 802.3cg, 10BASE-T1L) gives PROFIBUS field devices a direct path to Ethernet-class speed by replacing the physical medium itself rather than gatewaying around it: it carries Ethernet-standard protocols, including PROFINET, over two-wire cabling rated for severe conditions, with simultaneous power and data delivery, suitability for intrinsically safe hazardous areas, and bandwidth compatible with the Ethernet standard over cable runs up to 1000 m. In a deployed PROFINET network using this physical layer, measured application traffic reached 27.5 kbps in an 8-second sample, validating real operation while leaving most of the available bandwidth unused [31].

C. Technology Upgrading and Compatibility

Because equipment manufacturers continually release devices that support newer protocols, existing PROFIBUS systems must remain compatible or undergo costly and complex upgrades, while maintaining legacy operation alongside new technology adds further integration and maintenance complexity [6].

Improvement: Rather than building one-off adapters for each new device protocol, PROFINET functions as a structured integration platform: it defines concrete integration concepts (demonstrated using INTERBUS and PROFIBUS) that incorporate existing fieldbus technology into the real-time Ethernet network at a defined architectural level, achieving a high

degree of integration despite the differing device models of the legacy systems, supporting the gradual, investment-protecting introduction of new-protocol devices rather than forcing a wholesale replacement of PROFIBUS infrastructure. [32]

D. Physical-Layer Fragility

The RS-485 physical layer is highly sensitive to installation faults: excessively long cabling, missing active bus terminators, electromagnetic interference, and open- or short-circuit conditions degrade signal quality and communication performance, and such faults cause costly downtime [33].

Improvement: An improvement is automated physical-layer diagnosis using artificial neural networks that classify the signal waveform to identify the fault type; such a classifier can diagnose installation problems that disturb the waveform of signals transmitted on the PROFIBUS-DP physical layer, and with an appropriate feedback and retraining mechanism it can generalise to other network scenarios and settings, ensuring the viability of future installations [34].

E. Wired-Only Architecture and Limited Mobility

PROFIBUS is fundamentally a wired bus, which constrains deployment wherever mobility or hard-to-cable areas are involved. As the “last kilometer” of an Internet-plus-industrial-control network, PROFIBUS data increasingly needs to reach the Internet anytime and anywhere. [35]

Improvement: An improvement to remote/mobile connectivity is a dual-link gateway architecture: an OPC UA server on the wired side tunnels process data (flow rate, level, pressure, temperature, motor speed, etc.) to a remote OPC UA client over the internet, with two redundant links — fiber plus mobile data — so the second takes over if the first fails or degrades. In a practical deployment using this architecture, the remote client received link-failure/alarm notifications within 200–300 ms, while the local industrial network’s own real-time control loop never had to leave its wired connection. [36]

F. Absence of Built-in Cybersecurity

PROFIBUS was designed without authentication or encryption, which becomes a serious exposure as IT and OT networks converge: the lack of authorization and authentication allows a rogue device to connect and communicate on the bus and to access clear-text telegrams. In an attack-tree analysis of an industrial segment with a PROFIBUS backbone, the connected controller is a prime target because it can issue commands to disrupt the plant; with access to the master an attacker can also reach the slaves, read their data, reprogram them, map the topology, and spoof or capture traffic, a class of attack made easier by the absence of confidentiality controls and of mechanisms to validate connected masters [37].

Improvement: Because PROFIBUS-DP’s lack of authentication means any device with physical bus access can inject or manipulate traffic regardless of what security exists elsewhere in the network, a more targeted fix secures the bus segment itself through behavioral detection rather than perimeter controls. DpGuard automatically builds a finite-state

machine model of normal PROFIBUS-DP traffic from historical data, capturing legitimate state events, transitions, and transition probabilities across both the initialization and cyclic data-exchange phases, then flags real-time traffic violating those constraints as either a fault-injection or semantic attack. Deployed on a real PROFIBUS-DP system built from Siemens S7-300 and ET200 PLCs, it detected fault injection with 100% accuracy and semantic attacks with 99.80 % accuracy, adding only a microsecond-scale delay to bus communication. [38]

V. COMPARISONS

A. Profibus vs Profinet

PROFIBUS DP operates over RS-485 at baud rates from 9.6 kbit/s to 12 Mbit/s, using a master-slave polling model in which a DP master cyclically exchanges up to 244 bytes of I/O data with each slave [39]. A network can address up to 126 stations in total, with a maximum of 32 participants per RS-485 segment [40]. Determinism is achieved through the fixed polling sequence rather than a distributed clock: because the master controls every transaction, response times are bounded and calculable from the baud rate, slave count, and payload size [39]. Cycle time is therefore configuration-dependent; using Felser’s estimation formula, a 20-slave network at 1.5 Mbit/s yields approximately 5 ms, while a smaller configuration at 3 Mbit/s reaches 0.77 ms [41].

PROFINET, in contrast, is an Industrial Ethernet communication system. It serves a similar automation purpose as PROFIBUS, connecting controllers with field devices such as I/O modules, drives, and sensors, but it uses switched Ethernet instead of a serial bus. This gives PROFINET higher bandwidth, more flexible network topologies, easier integration with IT systems, and better diagnostic possibilities. For normal automation tasks, PROFINET RT is used for cyclic real-time data, while PROFINET IRT is available for applications that require stricter synchronization, such as motion control [42] [43].

B. Profibus vs ModBus

Modbus TCP is one of the earliest and most widely deployed industrial communication protocols, originating from the serial Modbus standard developed by Modicon in 1979 and later adapted for Ethernet by encapsulating Modbus function codes directly within TCP/IP frames [44]. Its primary strength lies in its simplicity and openness: the protocol is royalty-free, well-documented, and supported natively by virtually every industrial device manufacturer, making it a common choice for building automation, energy monitoring, SCADA systems, and legacy integration projects where multi-vendor interoperability is the priority [44].

In terms of real-time performance, Modbus TCP is fundamentally non-deterministic. It relies on explicit request-response polling over TCP/IP, meaning that cycle times and latency are directly affected by network congestion, switch load, and TCP stack overhead [45]. This places Modbus TCP outside the scope of hard real-time applications such as motion control or high-speed closed-loop systems, and firmly within

the domain of non-time-critical monitoring, configuration, and data acquisition tasks.

C. Profibus vs EtherNet/IP

EtherNet/IP (Ethernet Industrial Protocol) was developed by Rockwell Automation and is now maintained by ODVA. Like PROFINET, it adapts standard Ethernet infrastructure for industrial use by adding an application layer, in this case the Common Industrial Protocol (CIP), on top of TCP/UDP/IP, making it fully compatible with existing Ethernet hardware and network infrastructure [44]. CIP is also shared by DeviceNet and ControlNet, which gives EtherNet/IP a degree of cross-protocol interoperability through shared device profiles and object libraries [44].

EtherNet/IP supports two messaging modes. Explicit messaging uses TCP and follows a request-response model suited to configuration, diagnostics, and non-time-critical data. Implicit messaging uses UDP and operates on a producer-consumer model with a configurable Requested Packet Interval (RPI), which defines the cyclic update rate per connection [46]. In terms of determinism, standard EtherNet/IP over TCP/IP is soft real-time, with performance dependent on network load and infrastructure. CIP Motion elevates this to hard real-time by removing the determinism requirement from the network and placing it in the devices themselves [47]. EtherNet/IP operates at 100 Mbit/s to 1 Gbit/s and sees its strongest adoption in North American discrete manufacturing, food and beverage, and automotive assembly applications [44].

VI. CONCLUSION

PROFIBUS remains a reliable and deterministic fieldbus technology, especially in existing industrial plants where stability, interoperability, and long service life are important. Its master-slave and token-based communication mechanisms provide predictable data exchange, while standardized device descriptions support multi-vendor integration. However, its limited bandwidth, shared-bus structure, and lack of built-in cybersecurity make it less suitable for modern Industry 4.0 systems that require high-speed communication, IT/OT integration, and advanced diagnostics. Compared with newer Industrial Ethernet protocols, PROFINET provides a practical migration path, Modbus TCP remains useful for simple monitoring, and EtherNet/IP is widely adopted in Ethernet-based automation systems. Overall, PROFIBUS should be viewed as a mature legacy technology rather than an obsolete one, with gradual migration through gateways, proxies, and hybrid architectures being the most practical approach for combining existing PROFIBUS infrastructure with modern Ethernet-based communication.

REFERENCES

- [1] L. Zhou, H. Liu, and Y. Zhang, "Evolution of distributed control systems in industrial automation," *International Journal of Control and Automation*, 2020. [Online]. Available: <https://doi.org/10.1177/0020294020927420>
- [2] PROFIBUS and PROFINET International (PI), "Profibus system description," 2020. [Online]. Available: https://www.profibus.com/fileadmin/media/downloadsection/PROFIBUS_Systembeschreibung_ENG_web.pdf
- [3] PROFINET University, "Profibus pa – profinet transition in process automation." [Online]. Available: <https://profinetuniversity.com/process-automation/profibus-pa-profinet-transition/>
- [4] PI North America, "How does profibus dp work?" 2026, accessed: 2026-06-17. [Online]. Available: <https://us.profinet.com/how-does-profib-dp-work/>
- [5] "Performance analysis of profibus communication in real-time systems," *International Journal of Modern Engineering Research (IJMER)*, 2022. [Online]. Available: https://www.ijmer.com/papers/Vol15_issue4/15040511.pdf
- [6] Q. Zhe *et al.*, "Application status and challenges of profibus technology in industrial automation field," *International Journal of Mechanical Engineering and Education*, 2024. [Online]. Available: <https://doi.org/10.62051/ijmee.v2n2.03>
- [7] P. Aminaie, "Profinet vs profibus," 2020, arXiv:2011.14167. [Online]. Available: <https://arxiv.org/abs/2011.14167>
- [8] PROFINET International, "Profibus vs profinet," 2018. [Online]. Available: https://us.profinet.com/wp-content/uploads/2018/07/PB-vs-PN_v2.pdf
- [9] EtherCAT Technology Group, "Ethercat technology," accessed: 2026-06-17. [Online]. Available: <https://www.ethercat.org/en/technology.html>
- [10] pbMaster Project, "Profibus introduction." [Online]. Available: <https://sourceforge.net/p/pbmaster/wiki/profibus-introduction/>
- [11] Smowcode, "Profibus protocol: An in-depth exploration." [Online]. Available: <https://blog.smowcode.com/profibus-protocol-an-in-depth-exploration>
- [12] PROFINET (PI North America), "Profibus technology." [Online]. Available: <https://us.profinet.com/technology/profibus/>
- [13] Industrial Monitor Direct, "Profibus fms vs dp vs pa: Technical differences explained." [Online]. Available: <https://industrialmonitordirect.com/blogs/knowledgebase/profibus-fms-vs-dp-vs-pa-technical-differences-explained>
- [14] SAIABurgess Controls, "Profibus-fms manual." [Online]. Available: https://sbc-support.com/uploads/tx_srcproducts/26-742_EN_Manual_Profibus-FMS_01.pdf
- [15] AGH University of Science and Technology, "Profibus overview, course materials module 2." [Online]. Available: https://ipnet.agh.edu.pl/Materials1/Module2/ProfiBus_overview.pdf
- [16] Embien Technologies, "Profibus protocol: Communication profiles and physical layers." [Online]. Available: <https://www.embien.com/industrial-insights/profibus-protocol-communication-profiles-and-physical-layers>
- [17] J. P. M. De P. Paiva *et al.*, "Use case comparison between profibus pa and profinet pa networks," in *2025 IEEE 16th Latin America Symposium on Circuits and Systems (LASCAS)*, Feb. 2025, pp. 1–5.
- [18] Siemens AG, *Catching the Process Fieldbus: An Introduction to PROFIBUS for Process Automation*, 2nd ed. [Online]. Available: <https://assets.new.siemens.com/siemens/assets/api/uuid:3741da47-17f8-4d9c-b502-cc0da93df533/catching-the-process-fieldbus-2nd-edition.pdf>
- [19] AGH University of Science and Technology, "Profibus overview," Tech. Rep., accessed: Jun. 17, 2026. [Online]. Available: https://ipnet.agh.edu.pl/Materials1/Module2/ProfiBus_overview.pdf
- [20] G. Liang and X. An, "Simulation analysis of communication performance of profibus in single and multiple master mode," *Measurement and Control*, vol. 53, no. 7–8, pp. 1238–1249, Jun. 2020.
- [21] PROFIBUS & PROFINET International (PI), *PROFIBUS System Description: PROFIBUS Technology and Application*, 2016, accessed: Jun. 17, 2026. [Online]. Available: https://www.profibus.com/fileadmin/media/downloadsection/PROFIBUS_Systembeschreibung_ENG_web.pdf
- [22] PI North America, "Profibus," accessed: Jun. 17, 2026. [Online]. Available: <https://us.profinet.com/technology/profibus/>
- [23] K. Ali, R. S. Ambekar, V. V. Mehre, and S. Namekar, "Profibus dp & pa several issues in industrial network," in *Advances in Intelligent Systems and Computing*, 2024, pp. 303–318.
- [24] PROFINET University, "Profibus pa to profinet transition," accessed: Jun. 17, 2026. [Online]. Available: <https://profinetuniversity.com/process-automation/profibus-pa-profinet-transition/>
- [25] D. Krause, "Profibus fms vs dp vs pa: Technical differences explained," Apr. 2026, accessed: Jun. 17, 2026. [Online]. Available:

- <https://industrialmonitordirect.com/blogs/knowledgebase/profibus-fms-vs-dp-vs-pa-technical-differences-explained>
- [26] PI North America, "Marketing flyer: Profibus and profinet for process automation," accessed: Jun. 17, 2026. [Online]. Available: <https://us.profinet.com/documentation/marketing-flyer-profibus-for-process-automation/>
 - [27] —, "Process automation," Sep. 2012, accessed: Jun. 17, 2026. [Online]. Available: https://us.profinet.com/profibus_tech/process-automation/
 - [28] Editorial Staff, "How profibus communication works," Sep. 2018, accessed: Jun. 17, 2026. [Online]. Available: <https://instrumentationtools.com/how-profibus-communication-works/>
 - [29] Saia Burgess Controls (SBC), *Manual PROFIBUS-FMS*, accessed: Jun. 17, 2026. [Online]. Available: https://sbc-support.com/uploads/tx_srcprducts/26-742_EN_Manual_Profibus-FMS_01.pdf
 - [30] PBMaster, "Pbmaster / wiki / profibus-introduction," 2026, accessed: Jun. 17, 2026. [Online]. Available: <https://sourceforge.net/p/pbmaster/wiki/profibus-introduction/>
 - [31] A. B. Lugli, A. S. Aragão, E. R. Neto, G. A. M. Vizotto, J. A. P. Barbosa, J. P. M. M. Paiva, and T. C. Pimenta, "Survey and Practical Application of Ethernet-APL, PROFINET Network and HMI," Proc. 2025 Mixed Design of Integrated Circuits and Systems (MIXDES), Łódź, Poland, pp. 202–205, 2025.
 - [32] J. Jasperneite and J. Feld, "PROFINET: An Integration Platform for Heterogeneous Industrial Communication Systems," Proc. 10th IEEE Conf. on Emerging Technologies and Factory Automation (ETFA 2005), Catania, Italy, pp. 815–822, 2005.
 - [33] InstrumentationTools, "How PROFIBUS communication works," Online. [Online]. Available: <https://instrumentationtools.com/how-profibus-communication-works/>
 - [34] Industrial Monitor Direct, "PROFIBUS PA integration: PLC selection and gateway solutions," Online. [Online]. Available: <https://industrialmonitordirect.com/blogs/knowledgebase/profibus-pa-integration-plc-selection-and-gateway-solutions>
 - [35] PROFINET (PI North America), "Process automation," Online. [Online]. Available: https://us.profinet.com/profibus_tech/process-automation/
 - [36] M. B. Mahmood and J. M. Abdul-Jabbar, "Implementation of Dual Internet Links for Industrial IoT to Provide Safe Digital Commands for Process Automations," Journal Européen des Systèmes Automatisés (JESA), vol. 56, no. 1, 2023.
 - [37] pbMaster Project, "PROFIBUS introduction," SourceForge Wiki, Online. [Online]. Available: <https://sourceforge.net/p/pbmaster/wiki/profibus-introduction/>
 - [38] Z. Li, Q. Wei, R. Ma, Y. Geng, Y. Yang, and Z. Lv, "DpGuard: A Lightweight Attack Detection Method for an Industrial Bus Network," Electronics, vol. 12, no. 5, p. 1121, 2023. [Online]. Available: <https://www.mdpi.com/2079-9292/12/5/1121>
 - [39] PI North America, "How does PROFIBUS DP work?" Online, PROFIBUS & PROFINET International, accessed: Jun. 17, 2026. [Online]. Available: <https://us.profinet.com/how-does-profibus-dp-work/>
 - [40] K. Felsler, "Maximum number of stations," PROFIBUS Manual, Online, accessed: Jun. 17, 2026. [Online]. Available: https://felsler.ch/profibus-manual/maximale_anzahl_stationen.html
 - [41] —, "Estimation formula," PROFIBUS Manual, Online, accessed: Jun. 17, 2026. [Online]. Available: <https://www.felsler.ch/profibus-manual/schatzformel.html>
 - [42] PROFIBUS Nutzerorganisation e.V., "PROFINET system description: Technology and application," Online, PROFIBUS & PROFINET International, Nov. 2018, accessed: Jun. 17, 2026. [Online]. Available: https://www.profibus.com/fileadmin/media/downloadsection/technical_description_%26_books/PROFINET_System_Description_engl_2018_Update.pdf
 - [43] PI North America, "A complete comparison: PROFINET RT vs IRT," Online, PROFIBUS & PROFINET International, Apr. 2021, accessed: Jun. 17, 2026. [Online]. Available: <https://us.profinet.com/profinet-rt-vs-irt/>
 - [44] Industrial Monitor Direct, "Modbus TCP vs EtherNet/IP: Key protocol differences," Online, Mar. 2026, accessed: Jun. 17, 2026. [Online]. Available: <https://industrialmonitordirect.com/blogs/knowledgebase/modbus-tcp-vs-ethernetip-technical-protocol-comparison>
 - [45] W. Voss, "Industrial ethernet guide – technology and performance features," Copperhill Technologies, Online, Jul. 2019, accessed: Jun. 17, 2026. [Online]. Available: <https://copperhilltech.com/blog/industrial-ethernet-guide-technology-and-performance-features/>
 - [46] justprotocols.com, "EtherNet/IP protocol explained: CIP, scanner/adaptor model and implicit vs explicit messaging," Online, Mar. 2026, accessed: Jun. 17, 2026. [Online]. Available: <https://justprotocols.com/protocols/ethernet-ip>
 - [47] ODVA, "CIP motion," Online, accessed: Jun. 17, 2026. [Online]. Available: <https://www.odva.org/technology-standards/distinct-cip-services/cip-motion/>